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Climate-driven susceptibility of natural wildfires using Random Forest under future climate scenarios in Mediterranean forests of Türkiye

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Introduction: Mediterranean forest ecosystems are highly susceptible to natural wildfires under climate change, driven by rising temperatures, reduced precipitation, and pro¬longed dry periods. This study aimed to develop a climate-based natural wildfire susceptibility model using the Random Forest (RF) machine learning algorithm for the Emet Forest Management Directorate in the Mediterranean climate zone of Türkiye. This study considers only natural wildfires and excludes human-induced fire events.
Methods: All fire occurrence data used in the modeling process consist solely of recorded natural wildfire ignitions. The model incorporated 19 bioclimatic variables with historical wildfire occurrence data, using 258 recorded natural fire locations from 2015 to 2025 as reference points. Model performance was evaluated using Receiver Operating Characteristic (ROC) analysis.
Results and discussion: An Area Under the Curve (AUC) value of 0.711, which indicates moderate but acceptable predictive performance and is consistent with values reported in similar climate-driven susceptibility studies. The model results showed that the most influential drivers of fire susceptibility were temperature seasonality (BIO4), dry season precipitation (BIO17), and the minimum temperature of the coldest month (BIO6). According to future projections based on the Representative Concentration Pathways (RCP) 2.6 scenario using the Beijing Climate Center-Climate System Model Version 2-Medium Resolution (BCC-CSM2-MR) climate model for 2050 and 2070, the proportion of areas with high and extremely high fire susceptibility is projected to increase from 56.4% in 2025 to 64.0% in 2070, while low-and moderate-suscep¬tibility zones decline. This study provides one of the first climate-only, machine learning-based evaluations of present and future natural wildfire susceptibility in Mediterranean forests of Türkiye. These results revealed the escalating threat of wildfires in Mediterranean forests under climate change.

KEYWORDS
climate change scenarios, Mediterranean forests, Random Forest, ROC, wildfire susceptibility

1 Introduction

Wildfires, one of the most severe environmental disasters, arise from natural and anthropogenic processes and directly and indirectly disrupt ecosystem dynamics. These frequent global events cause serious structural and functional disruptions in forest ecosystems (Phillips et al., 2022). Annually, thousands of forest fires devastate over one million hectares of forested

areas, adversely affecting ecological stability and human safety, as well as socioeconomic development (Gao and Liao, 2017). From 2003 to 2012, forest fires destroyed an annual average of 67 million hectares of forests globally, representing roughly 1.7% of total forest resources (Kayet et al., 2020; Achu et al., 2021).

Lightning-ignited wildfires are caused by natural phenomena, including lightning strikes (Latham and Williams, 2001) and extreme temperatures (Hoover and Hanson, 2021). Global warming is altering the spatial and temporal patterns of wildfires worldwide by intensifying droughts, disrupting precipitation, and consequently increasing fire susceptibility and frequency (Abatzoglou and Williams, 2016; Bargali et al., 2023). The recent rise in wildfire susceptibility facilitates significant environmental degradation, jeopardizing ecosystem integrity (Bond and Keeley, 2005). Furthermore, wildfires degrade air quality by elevating particulate matter and greenhouse gas concentrations in the atmosphere, significantly devastate forest ecosystems, and accelerate global warming through carbon emissions (Bot and Borges, 2022; Wasserman and Mueller, 2023; Mishra et al., 2024).

Effective wildfire management and strategic forest sustainability measures are crucial for risk mitigation. Developing wildfire susceptibility maps is an essential and timely step, particularly for the safeguarding of Mediterranean forest ecosystems (Çolak and Sunar, 2020; Demir and Akay, 2024; Uçar et al., 2025). These maps provide critical information to forest managers, decision-makers, and firefighting teams by systematically evaluating environmental and human elements that may influence fire ignition. They are widely used by fire management organizations globally and are recognized as key tools for devising fire prevention and suppression strategies (Sivrikaya et al., 2014). Given the escalating threat, accurate and consistent national-scale susceptibility mapping is crucial for the effective management of fire-prone forests (Gibson et al., 2020). These maps facilitate understanding of the relationships between fire behavior and environmental parameters such as climate, topography, and vegetation, providing a structured basis for strategic planning to mitigate fire impacts (Keeley and Syphard, 2016). The resulting data enable land and forest managers to predict and categorize fire intensity, identify high-susceptibility zones, and formulate targeted intervention measures (Mitsopoulos et al., 2016).

Various statistical, spatial, and machine learning (ML) modeling approaches are employed to develop wildfire susceptibility maps (Rodrigues et al., 2014; Bektaş et al., 2025). These models conduct a spatial analysis of environmental, topographic, and anthropogenic factors to identify priority fire susceptible zones (Adab et al., 2013). In wildfire susceptibility analysis, ML models are highly effective owing to their exceptional ability to identify intricate patterns and interactions within multivariate data (Rihan et al., 2023; Shi and Zhang, 2023). These models are particularly effective at capturing spatial and causal linkages that traditional statistical methods often struggle (Massada et al., 2013). Commonly used ML algorithms include Support Vector Machines (SVM) (Kalantar et al., 2020; Banerjee, 2021; Piralilou et al., 2022), RF (Jain et al., 2020; Arjasakusuma et al., 2022; Trucchia et al., 2022; Yue et al., 2023), Artificial Neural Networks (ANN) (Ghorbanzadeh et al., 2019; Gholamnia et al., 2020), and Maximum Entropy (MaxEnt) (Yang et al., 2021; Sari, 2023; Sivrikaya and Demirel, 2025). These tools significantly assist decision-makers in establishing early warning systems (Elith et al., 2011; Oliveira et al., 2012; Pham et al., 2021; Zhang G. et al., 2019; Zhang Y. et al., 2019). In comparative studies highlighting RF's efficacy, Shao et al. (2022) reported that RF outperformed SVM and other ML methods in modeling wildfire susceptibility in China, and Tariq et al. (2022) demonstrated RF's superior accuracy compared with MaxEnt for wildfire susceptibility assessment. Riaño and Chuvieco (2021) demonstrated that RF-based models can comprehensively integrate climate, land use, and vegetation data for Mediterranean forests.

Atalay et al. (2024) employed the Canadian Forest Fire Weather Index (FWI) to assess 26 forest fires in Antalya that impacted over 50 hectares from 2000 to 2023. The integration of FWI values from local meteorological stations with satellite-derived FWI datasets at multiple spatial resolutions generated fire threat maps in their research. 85.7% of the burned areas were categorized as having high to extreme fire risk, indicating a strong correlation between elevated FWI values and major fires.

Ersoy et al. (2025) enhanced fire danger classification by integrating Geographic Information Systems (GIS), a literature-based model, the Analytic Hierarchy Process (AHP), and machine learning (ML) to develop the Forest Fire Danger Assessment System (FoFiDAS) for the Manavgat district of Antalya, Türkiye. Both modeling methodologies employed 13 parameters found in the literature. A comparative model analysis revealed a 53% regional overlap in fire threat classifications. The expert-weighted evaluation-based AHP offers a more systematic and locally appropriate classification methodology. The literature-based technique, although it employed more scientific evidence, raised concerns about contextual flexibility and site-specific sensitivity.

Tonbul (2024) developed a framework that integrates machine learning algorithms and understandable artificial intelligence to assess forest fire risks in the Mediterranean provinces of Mersin, Antalya, and Muğla in Turkey. The research employed three machine learning algorithms-XGBoost, GBM, and LightGBM- to generate FFS maps utilizing 14 fire conditioning elements, encompassing topographic, meteorological, environmental, and anthropogenic variables. The LightGBM model exhibits the best predictive accuracy at 0.897, followed by GBM at 0.881 and XGBoost at 0.851. The SHapley Additive Explanations (SHAP) method was employed to enhance model interpretability and emphasize the relative contributions of conditioning variables on forest fire vulnerability.

Tonbul et al. (2022) employed Sentinel-2A imagery to assess burn intensity in two forest fires that occurred in İzmir Province, the Aegean region of Türkiye, in 2017 and 2019. The research included pixel-based and object-based classification techniques, along with three machine learning algorithms for burn severity classification: Random Forest (RF), Rotation Forest (RotFor), and Canonical Correlation Forest (CCF). The CCF classifier outperformed RF and RotFor by 5% in both classification methods. The integration of ensemble-based classifiers with the differenced Normalized Burn Ratio (dNBR) generated burn severity maps.

RF models integrated with the Coupled Model Intercomparison Project Phase 6 (CMIP6) or comparable climate projections forecast future fire susceptibilities based on various climate change scenarios, thereby facilitating the assessment of climate change impacts and the formulation of adaptive fire management strategies (Parente et al., 2018; Rodrigues and de la Riva, 2021). The RF's variable importance ranking quantifies the relative influence of meteorological drivers on fire, enabling decision makers to formulate targeted intervention plans (Mohajane et al., 2021). RF-based wildfire susceptibility maps, especially when combined with climate projections, are essential for revealing the geographic impacts of climate change on fire regimes. Consequently, it facilitates the formulation of scientifically based assessments to identify future highly susceptible regions, guide forest

management strategies, optimize firefighting activities, and promote the sustainable management of natural resources (Bahadori et al., 2023).

Environmental and climatic factors that control fuel moisture, plant growth, and the overall flammability of an ecosystem are what cause most natural wildfires. The fire environment concept posits that wildfires result from a combination of fuels, weather, and ignition sources, with climate serving as the primary controlling factor at regional and continental levels (Keeley and Syphard, 2016). Temperature fluctuations, seasonal rainfall trends, and extended drought periods specifically affect fuel moisture levels and establish favorable conditions for natural fire ignitions and their subsequent spread (Abatzoglou and Williams, 2016). In this context, recent research highlighted that climate change could substantially modify the spatial distribution and frequency of natural wildfires by intensifying fire-prone environmental conditions (Pérez-Invernón et al., 2023). Consequently, this study emphasizes climate-induced vulnerability to natural wildfires instead of explicitly modeling particular ignition mechanisms, such as lightning probability or anthropogenic ignition processes.

Despite the abundance of research on fires in the Mediterranean region, investigations specifically addressing natural forest fires are scarce, and there is a notable absence of studies including climate change estimates. This study aims to assess the effects of climate change on natural wildfire susceptibility by identifying and mapping areas at risk of forest fires. In this context, bioclimatic variables were integrated with future climate projections using a machine learning algorithm, Random Forest (RF). This study is one of the first to produce both current and future (2050 and 2070) spatial fire susceptibility maps for the Aegan region of Türkiye, located in the Mediterranean climate zone, using a Random Forest model based solely on bioclimatic variables. The study integrates the BCC-CSM2-MR climate model and the RCP 2.6 and RCP 8.5 scenarios within the CMIP6 framework at high spatial resolution, demonstrating in detail the effects of these scenarios on fire susceptibility at the regional scale. Furthermore, the interpretation of the model's determining variables in relation to climatic dynamics makes a significant contribution to the spatial assessment of fire-climate interactions. In these respects, the study provides a unique scientific framework for the Mediterranean climate region, both in its methodological approach and in the climate-based projections it produces.

2 Materials and methods

2.1 Study area

This study was conducted within the Emet Forest Management Directorate (FMD) in the city of Kütahya, located in the Aegean Region of Türkiye. Emet FMD was selected as the study location due to the elevated danger of forest fires attributed to the abundance of fire-sensitive tree species and the prevailing meteorological conditions. The Emet FMD, covering 175,089.90 hectares, encompasses the districts of Emet, Çavdarhisar, and Hisarcık (Figure 1). Forest land constitutes 101,500.70 hectares (58%), while the remaining 73,589.20 hectares (42%) comprise non-forest cover, including agricultural land, pastures, settlements, and other open areas. The climate exhibits characteristics of a transitional zone, blending Mediterranean climate, continental, and Marmara influences, resulting in hot, dry summers and cold, snowy winters. The region's topography and climatic variability support a diverse forest ecosystem. Naturally occurring tree species forming both pure and mixed stands across various altitudinal zones include *Pinus nigra* Arnold., *Pinus brutia* Ten., *Juniperus* spp., *Quercus* spp., *Cedrus libani* A. Rich., and *Populus* spp.

2.2 Data description

Natural wildfires are significantly associated with climatic and meteorological circumstances. Climatic factors, including extended drought, elevated air temperatures, diminished relative humidity, and vigorous winds, decrease the moisture content of forest fuel, thereby enhancing its flammability and increasing the likelihood that natural wildfires will ignite and propagate. The spatial and temporal distribution of natural fires is intricately connected to climate conditions. Global research suggests that the escalating impacts of climate change are associated with increased natural wildfire risk and frequency in forests (Pérez-Invernón et al., 2023; Song et al., 2024). In this study, a static-input modeling framework was adopted. This study utilized 19 bioclimatic (temperature and precipitation) variables, representing annual trends, seasonality, and extreme climatic factors, which were obtained from the website[1] of WorldClim v2.1 (WorldClim, 2023). WorldClim v2.1 represents long-term climatological averages (baseline period 1970–2000) at a spatial resolution of 30 arc-seconds (about 1 km). Each bioclimatic variable was represented as a spatially continuous raster layer, from which a singular value per variable was retrieved for each pixel within the study area. The descriptions of temperature and precipitation variables are presented in Table 1. They are the standard set used in species distribution modeling and ecological studies. Forest fires in Türkiye are documented by the General Directorate of Forestry. Data on forest fires, including the date, time, location, coordinates, cause, and extent of the affected land, are documented. This study utilizes data from 258 natural forest fires in the Emet Forest Management Directorate between 2015 and 2025. Future bioclimatic variables for 2050 and 2070 were also obtained from WorldClim v2.1, based on the BCC-CSM2-MR global climate model under the low-emission scenario (RCP 2.6) and worst-case climate change scenarios (RCP 8.5).

2.3 Parameter selection

A variety of factors are employed in the development of wildfire susceptibility maps. Analysis of wildfire susceptibility studies reveals that bioclimatic variables are critical drivers of wildfire incidence, especially under climate change, and are therefore fundamental to spatial wildfire susceptibility maps (Sivrikaya and Küçük, 2022; Sivrikaya and Demirel, 2025). Natural forest fires are closely correlated with climatic and meteorological variables. This analysis focuses on bioclimatic variables related to natural forest fires, excluding topographic, demographic, and stand features. While this study initially considered all 19 bioclimatic variables, the presence of many correlated predictors is undesirable because it may lead to model overfitting and increased estimation uncertainty (De Marco and Nóbrega, 2018;

1 https://worldclim.org

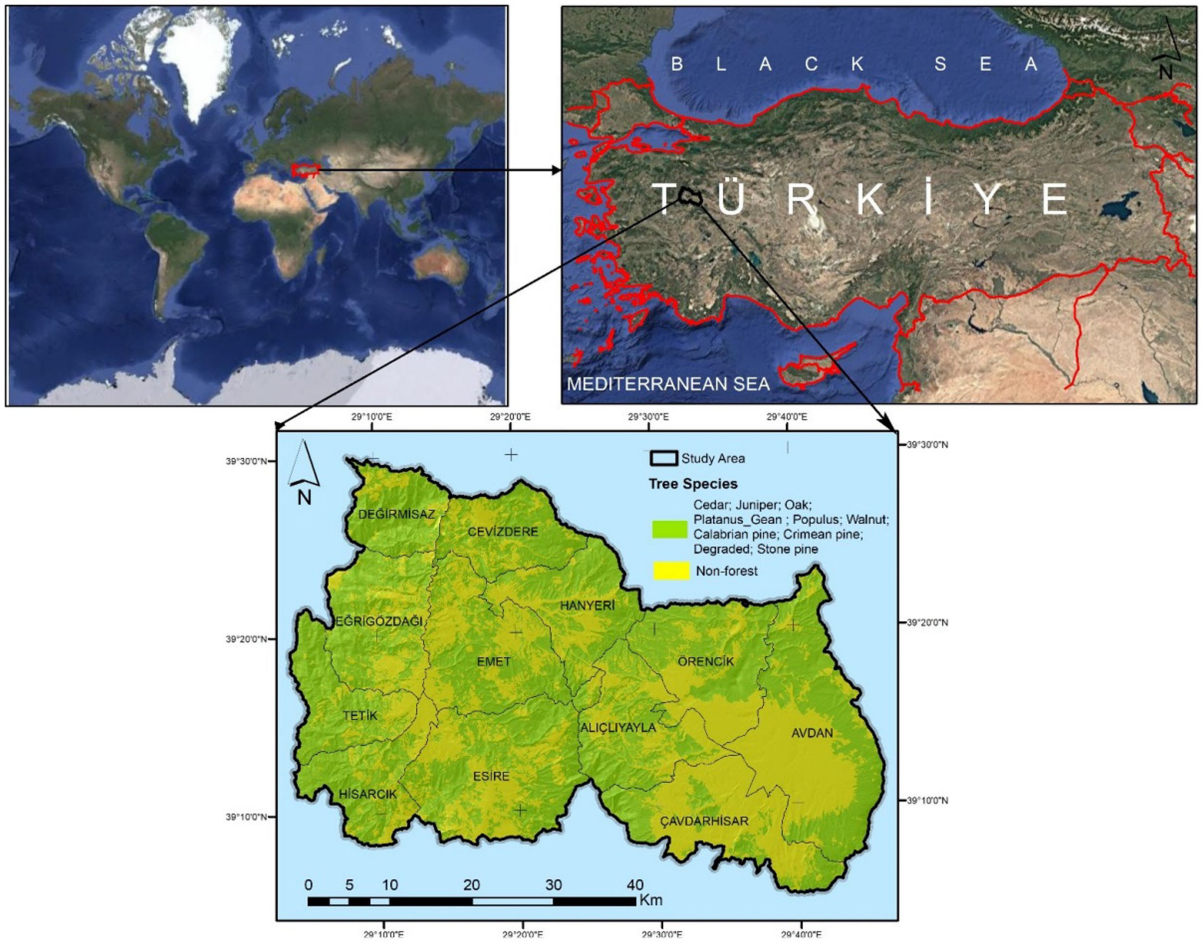


FIGURE 1
Study area covering the Emet Forest Management Directorate within the city of Kütahya.

Méndez-Encina et al., 2021; Özcan et al., 2025). A Pearson correlation analysis was conducted to mitigate these potential issues and select the most suitable predictors from a set of 19 variables. When two variables exhibited a strong correlation ($r > 0.9$), one of them was excluded from the model to reduce redundancy and enhance model stability (Yan et al., 2020; Sivrikaya and Demirel, 2025). This process ensured a more reliable selection of input variables for the subsequent analysis.

2.4 Random Forest algorithm

The Random Forest (RF) algorithm, developed by Breiman (2001), is a non-parametric machine learning method widely used for both classification and regression analysis. RF is an ensemble technique composed of multiple decision trees, each trained on a randomly selected subset of the data. Final predictions are derived through majority voting for classification problems and averaging methods for regression applications (Breiman, 2001; Mohajane et al., 2021; Hang et al., 2024). It is a non-parametric ensemble machine learning method widely used to predict wildfire risk because it shows how environmental factors can have nonlinear effects on fire behavior. We chose Random Forest over other statistical and machine learning methods, such as logistic regression or single decision trees, because it handles multicollinearity, outliers, and noisy data better. Also, it does a good job of determining the importance of different variables in large environmental datasets commonly used in wildfire research (Rodriguez-Galiano et al., 2012; Mohajane et al., 2021).

The RF model in this research was formulated using 500 decision trees generated via bootstrap aggregation. At each node, a random subset of predictor variables was chosen for splitting (mtry = $\sqrt{p}$), where p is the total number of input variables. To distinguish between scenarios likely to start a fire and those that aren't, the minimum terminal node size was set to 1. We chose these parameter values after reviewing other studies on wildfire risk and conducting initial tuning tests to reduce classification error and stabilize the model.

The input dataset included both examples of wildfires that occurred and those that did not in certain places. The data came from raster layers with a consistent spatial resolution (nearly 1 km grid cells). This is the usual way to map areas likely to catch fire. Fire occurrence samples were derived from historical wildfire records, while non-fire samples were randomly chosen from areas lacking documented fires to maintain environmental representativeness. To reduce sampling bias, we ensured that the amounts of fire and non-fire data were equal. Before modeling, all predictor variables were resampled and combined to ensure they had the same spatial resolution and projection.

We used 10-fold cross-validation to assess the model's performance. The dataset was split into training and validation sets at random. This process ensured that the model's ability to generalize

TABLE 1 Descriptions of temperature variables (WorldClim, 2023).

Code	Name	Description	Ecological relevance for fire
BIO1	Annual Mean Temperature	The average of all monthly mean temperatures.	A general indicator of thermal energy; higher values increase evapotranspiration and fuel drying.
BIO2	Mean Diurnal Range	The mean of the monthly (max temp − min temp).	Large ranges indicate high daytime heating and nighttime cooling, affecting daily fuel moisture cycles.
BIO3	Isothermality	(BIO2/BIO7) × 100. Ratio of diurnal to annual temperature range.	High values suggest a stable daily temperature regime relative to the annual range.
BIO4	Temperature Seasonality	Standard deviation of monthly mean temperatures × 100. Measures annual temperature variability.	Critical driver. High seasonality (hot summers, cold winters) creates pronounced dry/hot fire seasons.
BIO5	Max Temperature of Warmest Month	The highest monthly maximum temperature.	Direct fire driver. Peak heat stress, extreme fuel drying, and high ignition potential.
BIO6	Min Temperature of Coldest Month	The lowest monthly minimum temperature.	Warmer values can shorten winter dormancy, extend the fire season, and affect pest survival.
BIO7	Temperature Annual Range	BIO5 − BIO6. The difference between max of the warmest month and min of the coldest month.	Large ranges correlate with high continentality and intense summer fire potential.
BIO8	Mean Temperature of Wettest Quarter	Average temperature during the wettest 3-month period.	Indicates whether the wet season is warm (promoting fuel growth) or cool.
BIO9	Mean Temperature of Driest Quarter	Average temperature during the driest 3-month period.	Key fire variable. Hot dry seasons dramatically increase fire susceptibility by desiccating fuels.
BIO10	Mean Temperature of Warmest Quarter	Average temperature during the warmest 3-month period.	Indicates sustained seasonal heat stress on fuel moisture.
BIO11	Mean Temperature of Coldest Quarter	Average temperature during the coldest 3-month period.	Defines the low-energy period for plant growth and fire activity.
BIO12	Annual Precipitation	Total annual cumulative precipitation.	General water availability; low values indicate arid conditions conducive to fire.
BIO13	Precipitation of Wettest Month	Total precipitation in the single wettest month.	Extreme rainfall can indirectly affect soil moisture and fuel loads.
BIO14	Precipitation of Driest Month	Total precipitation in the single driest month.	Critical fire variable. Severe aridity indicates an intense monthly fuel moisture deficit.
BIO15	Precipitation Seasonality	Coefficient of Variation (CV) of monthly precipitation. Measures relative variation in rainfall.	High seasonality (distinct wet/dry periods) creates predictable annual fire seasons.
BIO16	Precipitation of the Wettest Quarter	Total precipitation during the wettest 3-month period.	Determines the peak growing season and the production of fine, flammable fuels.
BIO17	Precipitation of the Driest Quarter	Total precipitation during the driest 3-month period.	One of the most direct fire drivers. Defines the duration and intensity of the climatic dry season.
BIO18	Precipitation of Warmest Quarter	Total precipitation during the warmest 3-month period.	Crucial interaction. Warm + wet = fuel growth; warm + dry = extreme fire hazard.
BIO19	Precipitation of Coldest Quarter	Total precipitation during the coldest 3-month period.	Off-season precipitation recharges soil moisture, affecting next season's fuel moisture.

and make accurate predictions was carefully evaluated. After training the model, the RF method was applied to all grid cells in the research area to produce a continuous map of wildfire likelihood.

2.5 Variable importance measures

In the RF algorithm, variable importance is assessed based on its influence on the model's decision-making process. In this study, importance was measured using two commonly used methods: permutation importance (Mean Decrease in Accuracy) and Gini importance (Mean Decrease in Gini) (Breiman et al., 1984). A large reduction signifies the variable's increased significance to the model. Permutation importance (PI) measures the reduction in model accuracy resulting from randomly permuting each variable. The reduction in model accuracy when the values of a variable X_j are randomly permuted is calculated as (Equation 1):

$$\mathrm{PI}\left(X_j\right) = \mathrm{Accuracy}_{\mathrm{baseline}} - \mathrm{Accuracy}_{\mathrm{permuted}\left(X_j\right)} \quad (1)$$

Gini Importance (GI) measures how often a variable is used to make decisions in a model. The Gini impurity function quantifies the information gain provided by each variable at

each decision tree branch (Nembrini et al., 2018). Gini importance is computed by summing the Gini impurity decreases produced by splits on that variable across all nodes of all trees in the forest (Equations 2, 3). The total Gini importance (GI_{RF}) of variable X_j in a forest of M trees is:

$$GI_{RF}(X_j) = \frac{1}{M} \sum_{m=1}^{M} \sum_{t \in T_{j,m}} \Delta G_m(t, X_j) \quad (2)$$

$$\Delta G(t, X_j) = G(t) - \frac{N_L}{N_t} G(L) - \frac{N_R}{N_t} G(R) \quad (3)$$

ΔG = The reduction in Gini impurity.
$G(t)$ = Gini impurity at the parent node t.
$G(L)$, $G(R)$ = Gini impurity of left and right child nodes.
N_t = number of samples at node t.
N_L, N_R = number of samples in left and right child nodes.

These importance scores are aggregated across all trees and standardized so that the most influential variable is assigned a relative importance of 1. This normalization facilitates clear comparison among predictors and helps identify the factors that most strongly influence wildfire susceptibility classification (Rodriguez-Galiano et al., 2012).

2.6 Model implementation and evaluation

The RF model was implemented using the randomForest package in R v4.3.1. The dataset was randomly split into 70% for model training and 30% for independent validation. The model output is a continuous wildfire susceptibility index, with raster pixel values ranging from 0 (low susceptibility) to 1 (high susceptibility). This index raster was classified into four susceptibility categories (low, moderate, high, and extreme) using Jenks' natural breaks classification method in ArcGIS 10.6 to generate fire susceptibility maps. To generate the present susceptibility map, this index was calculated by applying the model to the set of contemporary (2025) bioclimatic variable rasters. To produce projected susceptibility maps for 2050 and 2070, the same trained model was applied to the corresponding future bioclimatic rasters from the BCC-CSM2-MR model under the RCP 2.6 and RCP 8.5 scenarios.

The validation of wildfire susceptibility maps is a crucial stage in the modeling process. The accuracy of the wildfire susceptibility model was validated using a Receiver Operating Characteristic (ROC) curve. The Area Under the Curve (AUC) is a composite statistic derived from ROC curves and is commonly used in related studies to evaluate the accuracy of wildfire susceptibility maps (Ma et al., 2020; Milanović et al., 2020). A ROC curve plots the true positive rate (sensitivity) against the false positive rate (1-specificity) across all classification thresholds. AUC provides a single metric of overall model performance: a value of 0.5 indicates random predictions, and 1.0 indicates perfect discrimination. AUC values between 0.5 and 0.7 indicate low precision, values from 0.7 to 0.8 indicate acceptable precision, values from 0.8 to 0.9 indicate outstanding precision, and values over 0.9 indicate remarkable model precision (Sivrikaya and Demirel, 2025; Özcan et al., 2025).

3 Results

3.1 Parameter selection and variable importance

To prevent model overfitting and identify the most suitable variables, the correlation among 19 bioclimatic variables was examined using the Pearson correlation analysis (r). The analysis revealed high correlation ($r > 0.9$) for 15 variables (BIO2, BIO3, BIO5, BIO8, BIO9, BIO10, BIO11, BIO13, BIO14, BIO16, BIO18, BIO19), which were consequently excluded from the final model (Figure 2). The RF model was thus constructed using seven key bioclimatic predictors: BIO1, BIO4, BIO6, BIO7, BIO12, BIO15, and BIO17 to assess wildfire susceptibility.

A significant feature of the RF model is its capacity to quantify the impact of each variable on the overall reduction in forecast accuracy. The significance of variables in RF classification indicates which factors the model deems most essential. In this study, the importance of these variables was assessed using the permutation importance (Mean Decrease in Accuracy) and Gini importance (Mean Decrease in Gini) indices (Breiman et al., 1984; Nembrini et al., 2018) (Figure 3). The results identified temperature seasonality (BIO4) as the most influential driver of wildfire susceptibility. Precipitation of the driest quarter (BIO17) and the minimum temperature of the coldest month (BIO6) were also highly significant. Annual mean temperature (BIO1) and precipitation seasonality (BIO15) were moderately important, while annual temperature range (BIO7) and total annual precipitation (BIO12) contributed least to the model. The findings indicate that temperature variations and precipitation throughout the dry season are the primary climatic determinants of wildfire susceptibility.

3.2 Current wildfire susceptibility and model performance

The Random Forest model generated the wildfire susceptibility map for the present period (2025) was categorized into four categories: low, moderate, high, and extreme (Figure 4). Table 2 presents the areal distribution of the susceptibility categories. The spatial distribution reveals that 56.4% of the study area falls within the high (25.0%) and extreme (31.4%) susceptibility categories. Low and moderate susceptibility zones cover 19.9 and 23.7% of the area, respectively. The accuracy of the wildfire susceptibility map produced by the RF model was assessed by ROC analysis. The AUC was 0.711 following 10-fold cross-validation (Figure 5). The results suggest that the wildfire susceptibility map produced by the RF model has acceptable predictive performance.

3.3 Future wildfire susceptibility (2050 and 2070)

Future susceptibility maps for 2050 and 2070 were projected using the Random Forest model under the RCP 2.6 and RCP 8.5 scenarios using the BCC-CSM2-MR climate model (Figure 6). The results according to the RCP 2.6 indicated a clear spatial expansion of higher susceptibility categories over time, accompanied by a contraction of lower susceptibility zones (Table 3). By 2050, the combined high and extreme susceptibility area is projected to increase to

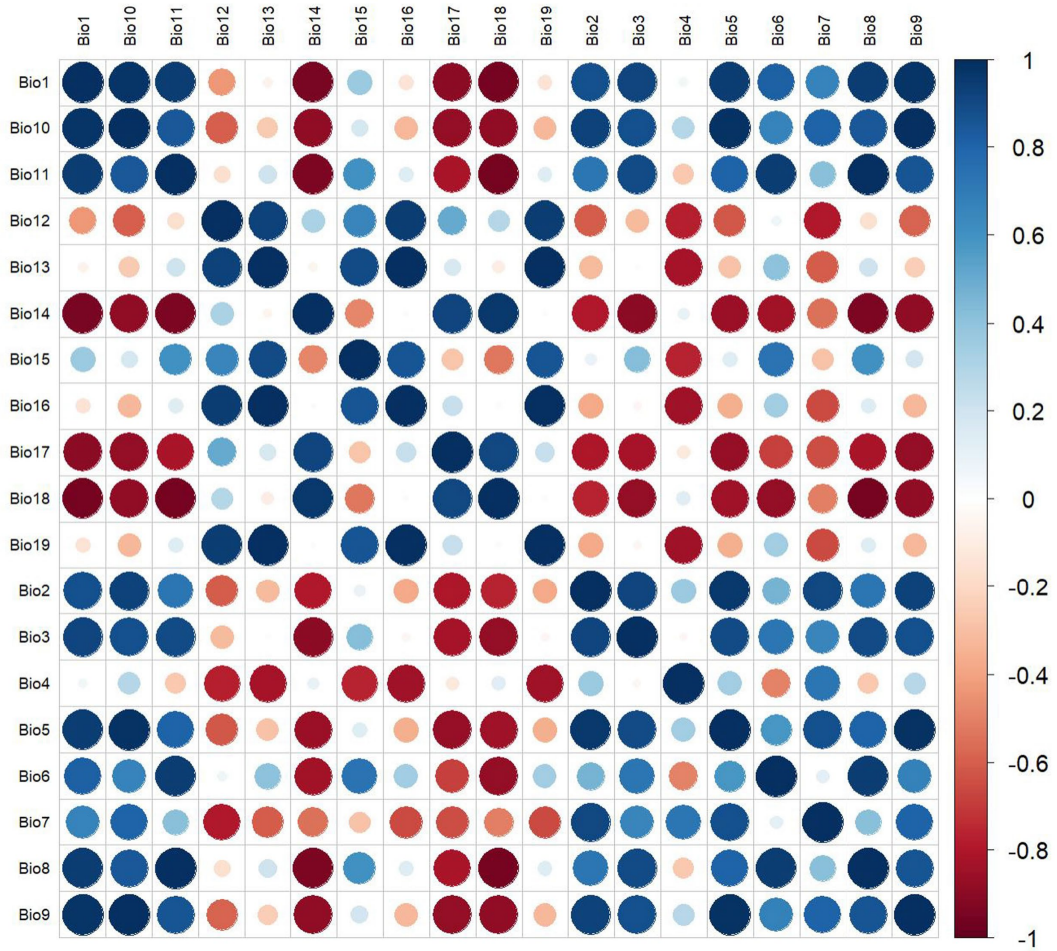


FIGURE 2
Correlation analysis of bioclimatic variables.

61.4% of the study region. This trend intensifies by 2070, with high and extreme zones expected to cover 64.0% of the area. Concurrently, low and moderate susceptibility areas are projected to decline from 43.6% in 2025 to 36.0% by 2070. The findings demonstrate that wildfire susceptibility is escalating under the modeled climate change scenario.

The RCP 8.5 scenario results indicate a more significant and rapid rise in fire susceptibility relative to the RCP 2.6 scenario. By 2070, it is anticipated that almost 70% of the study area will fall under the high and extreme susceptibility categories. The substantial reduction in low susceptibility areas suggests that the study area may evolve into a fire-prone area in the future (Table 3).

Projections for 2050 and 2070 indicate that transitions among fire susceptibility classes vary across scenarios. Under the RCP 8.5 scenario, regions with low and moderate susceptibility transition more swiftly to high and extreme susceptibility classifications. The shift from High to Extreme is more pronounced under RCP 8.5, with the Extreme region encompassing 50,460 hectares. Conversely, in the RCP 2.6 scenario, the transitions are more gradual and equitable, with the Extreme area expanding to 47,490 hectares by 2070. A reduction in Low and Moderate areas accompanied by a rise in High areas. Extreme conditions are evident in both scenarios; however, this change occurs more rapidly and is spatially more extensive in RCP 8.5. These data suggest that elevated emission circumstances increase fire susceptibility (Table 4).

4 Discussion

This study developed current (2025) and future (2050–2070) natural wildfire susceptibility maps for Emet FED located in the Mediterranean climate zone in Türkiye using an RF model based solely on bioclimatic variables. This approach directly assesses the impact of climate change on wildfire dynamics. The RF model achieved satisfactory predictive performance (AUC = 0.711), consistent with the accuracy levels typically observed in climate-driven wildfire susceptibility research using similar machine learning methodologies. This performance aligns with the typical AUC range (0.70–0.90) reported for RF-based wildfire susceptibility models in prior research (Guo et al., 2016; Ghorbanzadeh et al., 2019; Mitsopoulos et al., 2019; Shao et al., 2022; Trucchia et al., 2022).

The RF model has an adequate AUC (0.711), but the high proportion of high- and extreme-susceptibility areas (>56%) warrants careful interpretation. This pattern may be explained by the continuous susceptibility index and Jenks' natural breaks for categorical categorization, which favor relative spatial contrasts over absolute probability thresholds. Thus, high-sensitivity zones are meteorologically favorable for natural fires, not fire hotspots. Probability-based categorization algorithms or alternative thresholding methods may reduce overprediction and offer different areal proportions. Due to this constraint,

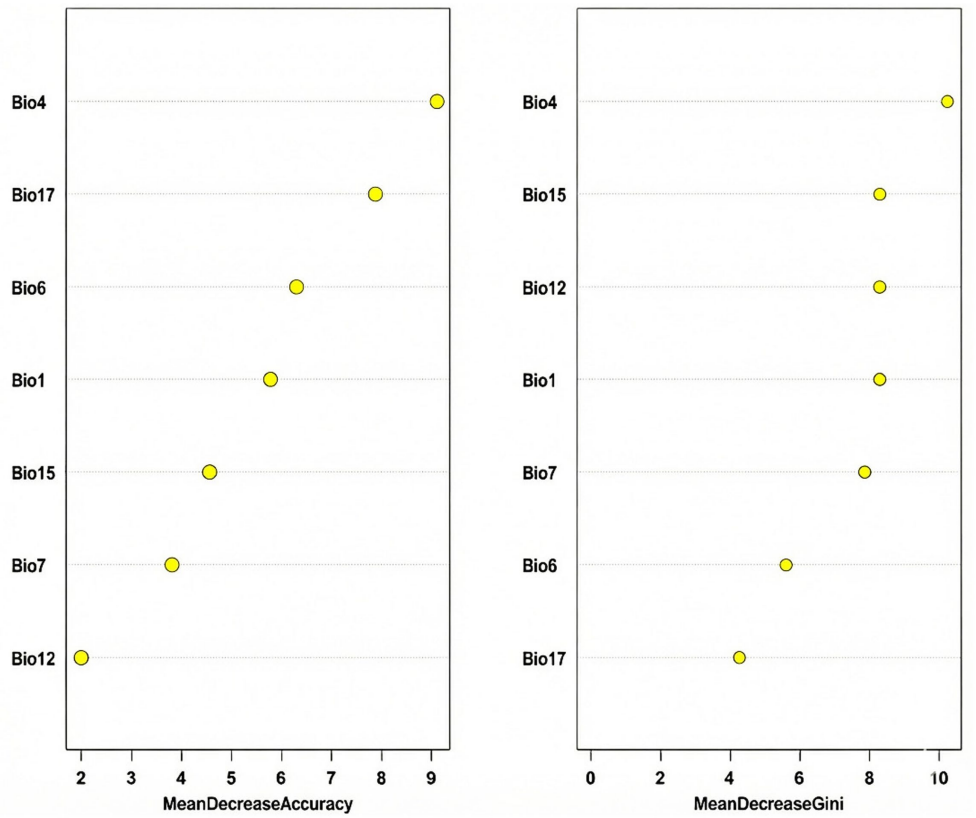


FIGURE 3
Mean decrease in accuracy and mean decrease in Gini index of selected bioclimatic variables.

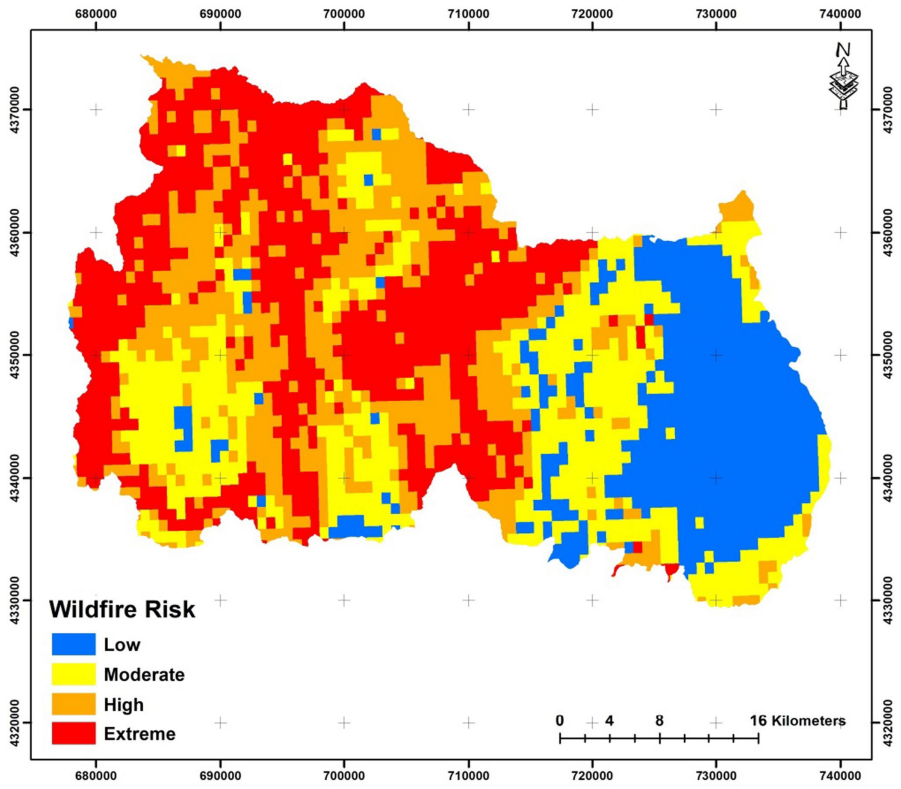


FIGURE 4
The wildfire susceptibility map is created using the Random Forest model, which categorizes wildfire risk levels.

strategic planning should employ susceptibility maps rather than deterministic projections.

The variable importance analysis revealed that temperature seasonality (BIO4), precipitation of the driest quarter (BIO17), and the minimum temperature of the coldest month (BIO6) were the most significant predictors in the RF model. These relevant rankings indicate the model's weather sensitivity, rather than the cause of wildfires. While temperature and precipitation variables may significantly correlate and influence significance ratings, climate-based modeling requires this distinction. Conclusions reflect the comparative model's efficacy rather than its causal claims. Despite these limitations, the variables show the meteorological factors influencing wildfire risk in the research area. Emet experiences temperature seasonality (BIO4) due to its Mediterranean-continental climate transition. Arid summers and cold winters influence fuel moisture and temperature in this region. Elevated temperature seasonality causes summer drought stress, accelerates fuel drying, and extends the fire season, making this area susceptible to wildfires.

Numerous comparative studies using RF, SVM, LR, and BRT algorithms found that RF is superior for generating wildfire susceptibility maps (Oliveira et al., 2012; Rodrigues et al., 2014; Shao et al., 2022).

TABLE 2 Distribution of the wildfire susceptibility categories for 2025.

Susceptibility categories	Area (ha)	%
Low	34,923.6	19.9
Moderate	41,560.9	23.7
High	43,844.8	25.0
Extreme	54,989.9	31.4

Other similar research has also confirmed RF's high performance relative to alternatives such as ANNs and Model Ensembles (ME) in evaluating wildfire susceptibility (Arpaci et al., 2014; Ghorbanzadeh et al., 2019). In a comparative assessment of nine algorithms (i.e., RF, ANN, SVM, radial basis function, least angle regression, self-organizing maps, and decision tree), Gholamnia et al. (2020) found the RF to be the most effective for generating wildfire susceptibility maps.

Nonetheless, it is important to note that other algorithms, including eXtreme Gradient Boosting (XGBoost), Maximum Entropy (MaxEnt), and ANN, have also demonstrated success across various research applications (Yang et al., 2021; Yue et al., 2023; Sari, 2024; Sivrikaya and Demirel, 2025). Shmuel and Heifetz (2022) compared RF, XGBoost, a multi-layer perceptron (MLP), and LR to assess a wildfire susceptibility map. It was reported that the XGBoost model was the most effective model for generating wildfire susceptibility maps. Yue et al. (2023) employed LR, ANN, RF, Gradient Boosted Decision Trees (GBDT), SVR, Light Gradient Boosting Machine (LGBM), XGBoost, and K-Nearest Neighbors (KNN) to assess wildfire susceptibility. It was found that the XGBoost model produced the best results. A separate study used XGBoost, RF, SVM, and Decision Tree (DT) to assess wildfire prediction efficacy, with the XGBoost model achieving the highest accuracy (Dong et al., 2022). In this respect, this work confirms that RF possesses robust predictive capability and produces effective outcomes in the Mediterranean climate zone. On the other hand, future evaluations of various models or the development of hybrid methodologies may enhance the precision of wildfire susceptibility assessment.

The primary variable in the model, assessed using climate parameters, was BIO4, which indicated temperature seasonality. This outcome further substantiates the critical influence of meteorological variables on fire possibility. Research indicates that temperature and dry-season precipitation are the primary determinants

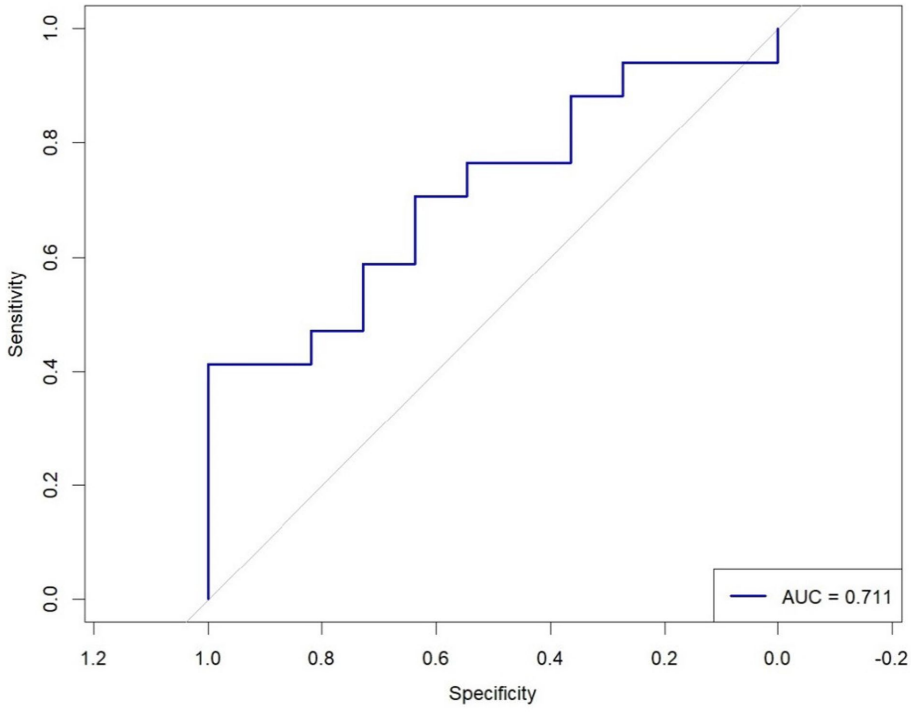


FIGURE 5
ROC curves for the wildfire susceptibility map produced by the Random Forest model.

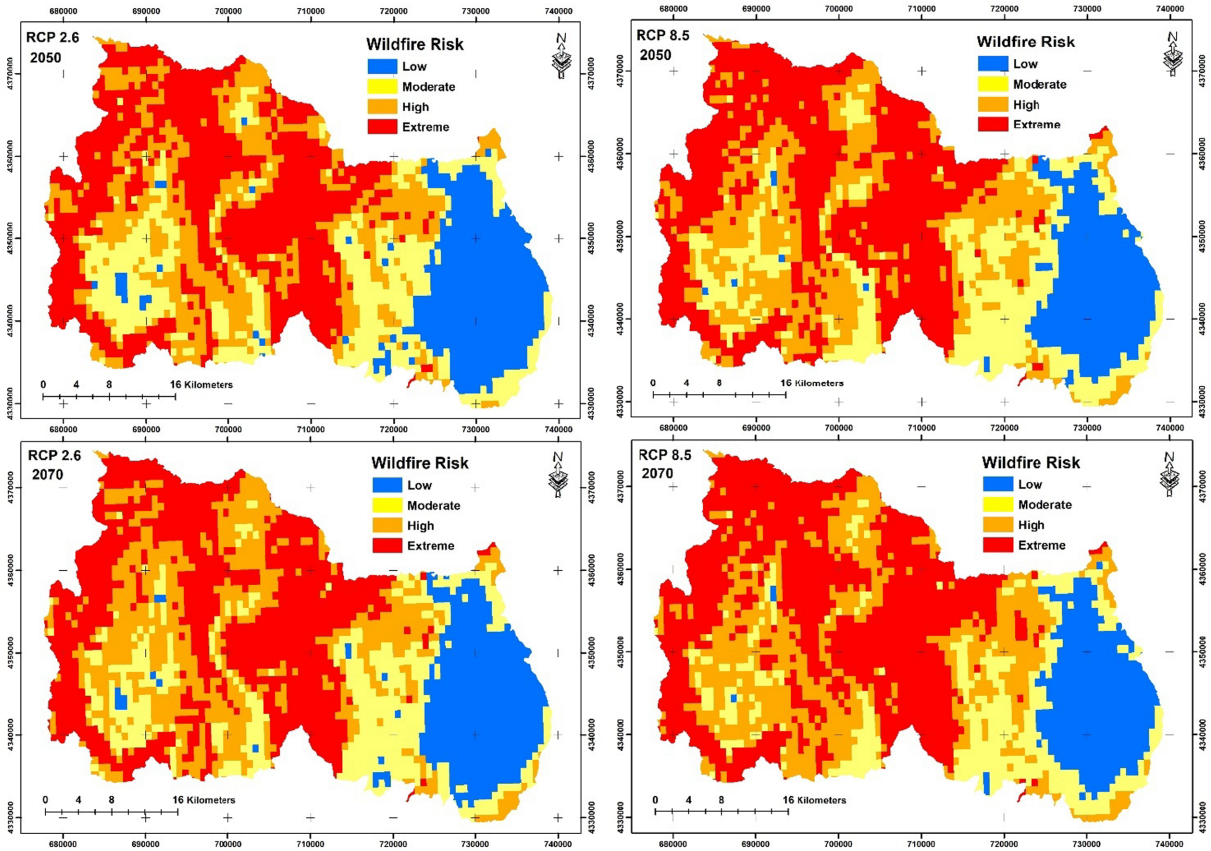


FIGURE 6
Wildfire susceptibility maps for the years 2050 (top left) and 2070 (bottom left) according to the RCP 2.6 scenario and 2050 (top right) and 2070 (bottom right) according to the RCP 8.5 scenario, based on the random forest model.

TABLE 3 Projected changes in wildfire susceptibility categories for 2050 and 2070 under the RCP 2.6 and RCP 8.5 scenarios.

Susceptibility categories	2025		RCP 2.6				RCP 8.5			
			2050		2070		2050		2070	
	Area (ha)	%	Area (ha)	%	Area (ha)	%	Area (ha)	%	Area (ha)	%
Low	34,923.6	19.9	31,450.2	17.9	28,873.9	16.5	25,197.3	14.4	24,318.8	13.9
Moderate	41,560.9	23.7	36,293.9	20.7	34,113.1	19.5	36,217.7	20.7	28,619.6	16.3
High	43,844.8	25.0	45,845.0	26.1	48,909.5	27.8	47,944.4	27.3	49,587.8	28.3
Extreme	54,989.9	31.4	61,730.1	35.3	63,422.7	36.2	65,959.8	37.6	72,793.0	41.5
Total	175,319.2	100.0	175,319.2	100.0	175,319.2	100.0	175,319.2	100.0	175,319.2	100.0

of wildfire susceptibility in Mediterranean ecosystems (Mitsopoulos et al., 2019). Abatzoglou and Williams (2016) asserted that rising global temperatures elevate both the frequency and intensity of wildfires. The results of our research on dry-season precipitation (BIO17) and minimum temperature (BIO6) indicate a robust correlation among fires, droughts, and heat waves. Moreover, research conducted by Oliveira et al. (2012) and Trucchia et al. (2022) revealed that climatic conditions are the primary determinants of fire dynamics. Shao et al. (2022) highlighted in their study using the RF model that seasonal variation in temperature and precipitation is a critical factor influencing wildfire susceptibility. Modeling studies often forecast a rise in both fire frequency and fire severity globally (Keeley and Syphard, 2016).

The spatial analysis revealed a concerning forecast; areas categorized as high and extreme fire susceptibility are projected to increase from 56.4% (2025) to 64.0% (2070, RCP 2.6 scenario) and to 70.0% (2070, RCP 8.5 scenario), while low- and moderate-susceptibility areas are projected to decrease. The results are consistent with projections for the Mediterranean ecosystem, where climate change is expected to increase fire weather conditions (Oliveira et al., 2012;

TABLE 4 Transition areas between wildfire susceptibility categories for 2050 and 2070 under the RCP 2.6 and RCP 8.5 scenarios.

Years	Susceptibility categories	Low	Moderate	High	Extreme
		RCP 2.6 Scenario			
		2050			
2025	Low	28,963.2	5,305.2	650.0	4.2
	Moderate	2,478.1	25,608.0	12,868.3	604.5
	High	6.9	5,221.2	23,455.0	15,158.7
	Extreme	1.0	157.5	8,868.7	45,958.7
		2070			
2025	Low	27,246.4	7,014.9	657.8	3.5
	Moderate	1,557.6	23,329.2	16,201.6	470.5
	High	68.5	3,574.8	24,744.1	15,454.4
	Extreme	0.4	192,0.2	7,303.1	47,490.2
		RCP 8.5 Scenario			
		2050			
2025	Low	24,263.3	9,990.9	667.9	0.5
	Moderate	931.3	23,548.2	16,766.1	313.3
	High	1.6	2,669.2	23,947.3	17,223.7
	Extreme	0.1	7.4	6,560.1	48,418.3
		2070			
2025	Low	23,649.6	10,129.3	1,141.8	1.9
	Moderate	666.7	17,479.9	22,013.4	1,398.9
	High	1.4	1,005.7	21,907.0	20,927.7
	Extreme	0.1	2.7	4,522.6	50,460.5

Trucchia et al., 2022). These forecasts show increased fire-prone areas, particularly in the Mediterranean basin, in line with future climate change projections (Keeley and Syphard, 2016). Similar forecasting studies also indicate a rise in highly susceptible regions beyond 2050 (Karali et al., 2014; Mitsopoulos et al., 2016; Rovithakis et al., 2022) due to rising temperatures identified as a key driver of this elevated susceptibility in the Mediterranean climate zone (Sivrikaya and Demirel, 2025). The findings indicate that climate-induced rises in temperature variability and dry-season water shortages are expected to intensify stress on forest ecosystems in the region in the forthcoming decades. The reported 13.48% rise in high and extreme vulnerability areas between 2025 and 2070 is scenario-dependent, not a quantitative forecast. This quantity depends on the climate model, the emission scenario, the predictor set, and the susceptibility categorization approach. One low-emission scenario and a fixed classification scheme may reduce or exaggerate the apparent rate of change. Thus, the projected rise should be seen as a consistent directional trend-the extension of climatically favorable wildfire conditions-rather than an accurate assessment of future areal change.

A well-established, widely recognized machine learning technique, wildfire susceptibility mapping employs Random Forests. In environmental and fire studies, RF is preferred for modeling nonlinear relationships, handling multicollinearity among predictors, and capturing intricate interactions in high-dimensional datasets. This study does not juxtapose RF with alternative machine learning or statistical approaches. Consequently, RF is introduced as an appropriate and resilient modeling framework for analyzing wildfire vulnerability influenced by regional climates rather than as an optimal strategy. The findings indicate that a climate-oriented Random Forest model can represent regional wildfire vulnerability trends and their anticipated changes under future climatic conditions. The model provides strategic insight into the meteorological aspects of wildfire risk, excluding anthropogenic, topographical, and fuel factors. Consequently, susceptibility maps can enhance long-term forest management, climate resilience, and the identification of areas requiring supplementary monitoring or proactive management due to heightened climatic stress. Explainable AI, or comparative modeling, may aid future climate studies in identifying variables. Projection maps should be regarded as scenario-based indicators of potential trends rather than definitive forecasts. Future investigations utilizing diverse climate models, emission trajectories, and non-stationary modeling frameworks will be essential for a more thorough evaluation of uncertainty.

5 Conclusion

This study assessed current and future wildfire susceptibility in the Emet Forest Management Directorate using a Random Forest model based on bioclimatic variables. The model demonstrated acceptable predictive accuracy (AUC = 0.711) and identified temperature seasonality (BIO4), dry season precipitation (BIO17), and the minimum temperature of the coldest month (BIO6) as the primary climatic determinants of fire susceptibility in this Mediterranean climate zone.

These findings are consistent with known climatic factors controlling fire regimes in Mediterranean ecosystems.

Projections under the RCP 2.6 scenario indicate a spatial trend: the total area of high and extreme susceptibility categories is projected to increase by 14% from 2025 to 2070, while lower-susceptibility areas are projected to decrease accordingly. The RCP 2.8 scenario also shows a comparable change, with a projected 24% increase in regions classified as having high or extreme sensitivity. These shifts are a clear indication of increasing climate-driven pressure on forests in the Mediterranean region.

The main contribution of this study is methodological: it demonstrates that a climate-based Random Forest model can produce spatially sensitive projections useful for long-term strategic planning. The generated maps can inform climate change adaptation frameworks such as fuel management, forest restoration, or the identification of priority areas for monitoring infrastructure.

It is important to note the study's limitations. The model relies solely on bioclimatic variables, excluding other critical factors such as topography, human activity, and fuel characteristics. This may limit the model's applicability to operational and short-term fire suppression activities. Furthermore, the projections depend on the specific climate model (BCC-CSM2-MR) and emissions scenario (RCP 2.6 and RCP 8.5) used. Therefore, while the results highlight the increasing threat of wildfire under climate change, they should be interpreted as complementary and strategic information rather than a standalone operational tool. Future research may integrate anthropogenic and topographic variables to improve local-scale accuracy and compare multiple climate models and emissions scenarios to better capture uncertainty in long-term projections.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author.

Author contributions

NB: Writing – original draft, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that Generative AI was not used in the creation of this manuscript.

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